MATHEMATICAL BIOLOGY

How Does a Hydra Decide Where to Grow a Head?

Chemical and mechanical routes to pattern formation

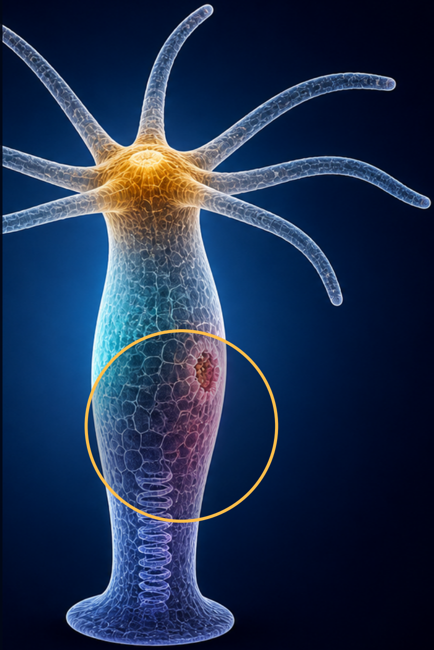
From regeneration to Turing instability, springs and energy

LECTURE MAP

The road from Hydra to pattern formation

From biological observations to mathematical mechanisms.

01 **Growth, wounds & long-range dynamics**



What do we mean by a pattern?

Return to the Gierer–Meinhardt system

$$\partial_t u = D_u \Delta u + \frac{a u^2}{K + v} - \mu_u u$$
$$\partial_t v = D_v \Delta v + b u^2 - \mu_v v$$

with homogeneous Neumann boundary conditions on Ω

STEADY STATE

$$0 = D_u \Delta \bar{u} + \frac{a \bar{u}^2}{K + \bar{v}} - \mu_u \bar{u}$$
$$0 = D_v \Delta \bar{v} + b \bar{u}^2 - \mu_v \bar{v}$$

- **Spatial pattern**
  A stable nonconstant steady state $(\bar{u}(x), \bar{v}(x))$

For nonlinear reaction–diffusion systems, a complete characterization of all steady states is generally not known

01 GROWTH, WOUNDS & LONG-RANGE DYNAMICS

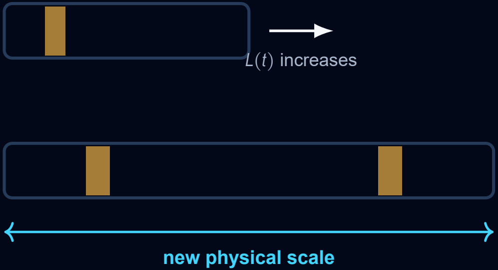
Growth and division

Growth and division change both the geometry and the initial data of the nonlinear problem

Growth makes the domain part of the dynamics

TIME-DEPENDENT DOMAIN

$$\Omega(t) = [0, L(t)]$$
$$L'(t) > 0$$



- **Changing geometry**
  Growth changes physical distances and the spatial structures that fit inside the domain
- **Nonlinear question**
  Does the current pattern persist, reorganize, or generate additional peaks

The concentrations evolve while the domain supporting them changes

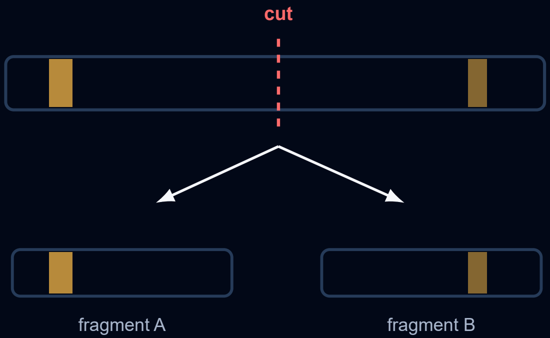
Growth can transform one stable pattern into another

a new spatial structure can form as the domain grows

the simulation follows the full nonlinear solution in the physical coordinate

Growth can reorganize an existing pattern instead of merely stretching it

Division creates two new initial-value problems



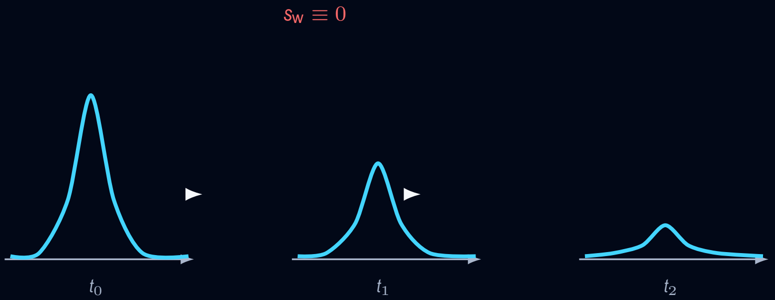
- **Inherited profiles**
  Each fragment receives the restriction of the pre-cut profiles of u and v
- **New boundaries**
  No-flux conditions are imposed at the newly created ends
- **Independent evolution**
  After division the fragments solve two separate nonlinear problems

01 GROWTH, WOUNDS & LONG-RANGE DYNAMICS

Role of the wound

Regeneration requires a localized wound signal strong enough to cross the nonlinear activation threshold

A cut alone does not guarantee regeneration



- **Simulation outcome**
  Without wound input the headless fragment can converge to $(0, 0)$
- **Nonlinear explanation**
  The inherited profile remains below the activation threshold and inside the basin of $(0, 0)$

A new boundary does not by itself move the system into the basin of a nonzero pattern

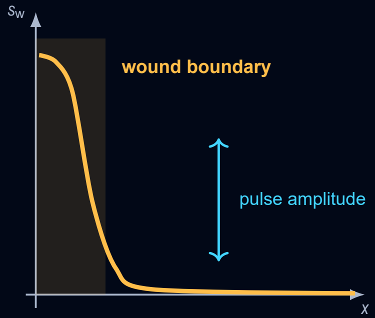
A wound provides a localized transient signal

ACTIVATOR EQUATION WITH WOUND INPUT

$$\partial_t u = D_u \Delta u + \frac{au^2}{K+v} - \mu_u u + s_\mathrm{w}(x,t)$$

$$\operatorname{supp} s_\mathrm{w} \subset [0, \ell_\mathrm{w}] \times [0, T_\mathrm{w}]$$

- **Localized in space**
  The perturbation acts near the newly created wound boundary
- **Transient in time**
  The input disappears after a finite activation period



- **Threshold crossing**
  Sufficient amplitude and duration move the local state across the activation threshold

A transient wound can create a persistent organizer

the wound input disappears while the newly initiated peak persists

after threshold crossing, the autonomous GM dynamics can approach a stable nonconstant steady state

The wound initiates regeneration while the nonlinear feedback maintains the resulting organizer

The simulations expose a missing biological mechanism

WHAT THE SIMULATIONS DEMONSTRATE

- **Several stable patterns**
  The full nonlinear system can support different steady spatial profiles
- **Geometry-dependent outcomes**
  Growth and division change the domain and the initial-value problem
- **Wound-dependent regeneration**
  A transient localized input can initiate a persistent organizer

WHAT THE MODEL STILL ASSUMES

- **Long-range molecular inhibition**
  The classical mechanism requires an inhibitor that spreads farther than the activator
- **Uncertain biological identity**
  Candidate activators exist, but the corresponding rapidly spreading inhibitor remains unclear

Could mechanics supply the missing inhibition?

This motivates a mechanochemical model in which tissue deformation participates in pattern formation